

Observable-specific Mpemba effect in double-well Langevin dynamics: role of metastable basin imbalance

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We demonstrate a robust Mpemba-like anomalous relaxation in overdamped Langevin dynamics with a double-well potential $U(x, y) = b(x^2 - 1)^2 + 0.3y^2$. In a systematic study of 720 independent simulations across 9 barrier heights and 8 temperatures, we find that the effect is **observable-specific**: a “hot” initial condition (spread across both wells) relaxes to equilibrium basin proportions faster than a “cold” condition (trapped in one well) for the basin-occupancy observable p_{left} , while no crossing occurs in energy or position variance. The effect is strong ($> 90\%$ of trajectories cross) in a well-defined region of parameter space ($b = 0.5\text{--}2.0$, $T = 0.1\text{--}0.2$), with a peak of 95.4% at $b = 2.0$, $T = 0.2$. Four independent controls confirm the mechanism: balanced-cold initialization eliminates the effect, single-well potential shows no effect, results are unchanged without gradient clipping, and 30 random seeds all produce strong crossings. Spectral analysis of the Fokker–Planck operator reveals that the Mpemba region coincides with large timescale separation between inter-basin (slow) and intra-basin (fast) relaxation modes. The Kramers escape rate formula predicts the slow eigenvalue with 89% accuracy (within factor 2). We argue that the observable-specificity arises because basin occupancy projects onto the slow inter-basin mode, while energy-like observables project predominantly onto fast intra-basin modes. Our results suggest that previous negative reports of the Mpemba effect may reflect the choice of observable rather than the absence of the phenomenon.

I. INTRODUCTION

The Mpemba effect—the counterintuitive observation that a hotter system can sometimes relax to equilibrium faster than a cooler one—has been a subject of renewed scientific interest following its rigorous formulation in Markovian dynamics [1–3]. Originally observed in water freezing [4], analogues have since been demonstrated in spin systems [7], colloidal particles [6], and granular gases [5].

Recent theoretical work has established that the Mpemba effect in Markov processes is governed by the spectral structure of the transition operator [1, 2]. Specifically, if the slowest relaxation mode has an anomalous initial overlap with the hot distribution compared to the cold one, faster relaxation can occur. Lu and Raz [1] provided a general spectral criterion, and subsequent work has explored strong vs. weak forms of the effect [3, 9]. A recent thermomajorization framework [10] proves that when initial states satisfy a thermomajorization ordering, the Mpemba effect holds for all monotone distance measures simultaneously—eliminating observable dependence entirely. Most recently, Summer et al. [14] provided a resource-theoretical unification of classical and quantum Mpemba effects, showing that both arise from the same underlying logic of athermality as a resource.

Despite this theoretical progress, a key question remains underexplored: **what happens when thermomajorization does not hold?** In that regime, different observables may couple to different relaxation modes,

and the Mpemba effect should depend on which mode dominates the observable of interest. Most studies define relaxation in terms of a global distance measure (total variation, KL divergence, or energy), leaving the observable-specific regime uncharted.

In this work, we address this question directly. We study overdamped Langevin dynamics in a 2D double-well potential and show that:

1. A robust Mpemba-like crossing occurs in the basin-occupancy observable p_{left} , with $> 90\%$ of trajectories crossing in the optimal parameter regime.
2. **No crossing occurs** in energy or position-variance observables for the same trajectories and parameters.
3. The mechanism is metastable basin imbalance: the hot distribution starts closer to the equilibrium basin proportions, while the cold distribution is trapped in one well.
4. The phase boundary of the effect is predicted by the timescale separation between the inter-basin and intra-basin relaxation modes of the Fokker–Planck operator.

These results demonstrate that the Mpemba effect is not a property of the system alone, but of the system-observable pair. This observable-specificity has direct implications for experimental design: searching for the Mpemba effect with the wrong observable may yield a false negative.

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II. MODEL AND METHODS

A. System

We consider overdamped Langevin dynamics in 2D:

$$d\mathbf{x} = -\nabla U(\mathbf{x}) dt + \sqrt{2T} d\mathbf{W} \quad (1)$$

with the double-well potential

$$U(x, y) = b(x^2 - 1)^2 + 0.3y^2 \quad (2)$$

where $b > 0$ is the barrier height and $T > 0$ is the temperature. The potential has two minima at $(x, y) = (\pm 1, 0)$ separated by a barrier of height b at the origin.

B. Initial conditions

We define two classes of initial conditions:

- **Cold:** particles sampled from $\mathcal{N}((1, 0), \sigma_c^2 \mathbf{I})$ with $\sigma_c = 0.05$. All particles start in the right well.
- **Hot:** particles sampled from $\mathcal{N}((0, 0), \sigma_h^2 \mathbf{I})$ with $\sigma_h = 2.0$. Particles are spread across both wells approximately equally.

C. Observables

We track three observables:

1. **Basin occupancy:** $p_{\text{left}}(t)$ = fraction of particles with $x < 0$.
2. **Mean energy:** $\langle E(t) \rangle = \langle U(x(t), y(t)) \rangle$.
3. **Position variance:** $\langle x(t)^2 \rangle$.

For each observable $O(t)$, we define the distance to equilibrium as $|O(t) - O_{\text{eq}}|$, where O_{eq} is estimated as the average of the final 10% of both trajectories.

D. Mpemba crossing criterion

A Mpemba-like crossing is detected when the hot trajectory's distance to equilibrium drops below the cold trajectory's distance:

$$|O_{\text{hot}}(t_c) - O_{\text{eq}}| < |O_{\text{cold}}(t_c) - O_{\text{eq}}| \quad (3)$$

for some time t_c , with the initial ordering reversed: $|O_{\text{hot}}(0) - O_{\text{eq}}| > |O_{\text{cold}}(0) - O_{\text{eq}}|$. We quantify the effect by the fraction of simulation time during which the hot trajectory is closer to equilibrium.

E. Simulation parameters

Each simulation uses $N = 5000$ particles with step size $\eta = 0.005$ and $n_{\text{steps}} = 30,000$ (evaluated every 50 steps). For the phase diagram, we sweep $b \in \{0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0, 5.0\}$ and $T \in \{0.05, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 1.0\}$, with 10 independent random seeds per point (720 runs total). Robustness is confirmed with 30 seeds at the reference point ($b = 1.0$, $T = 0.2$).

F. Spectral analysis

The Fokker–Planck operator for the 1D marginal (integrating out y) is transformed to Schrödinger form via the similarity transformation $H = e^{U/2T} \mathcal{L} e^{-U/2T}$, yielding a real symmetric operator with eigenvalues $0 = \lambda_0 > -\lambda_1 > -\lambda_2 > \dots$. The rates λ_k give the relaxation timescales $\tau_k = 1/\lambda_k$. We compute these numerically via sparse eigenvalue decomposition (shift-invert mode) on a grid of 501 points, and compare with the Kramers escape rate formula for λ_1 .

III. RESULTS

A. Observable-specific crossing

Figure 1 shows the central result. At the reference parameters ($b = 1.0$, $T = 0.2$), the distance to equilibrium for p_{left} shows a clear crossing: the hot trajectory relaxes faster and remains closer to equilibrium for $> 94\%$ of the simulation. In contrast, the energy and position-variance observables show no crossing—the hot trajectory remains farther from equilibrium throughout.

This observable-specificity is the key finding. The same trajectories, same dynamics, same initial conditions produce opposite conclusions depending on which observable is measured.

B. Mechanism: metastable basin imbalance

Three control experiments (Fig. 3) confirm the mechanism:

(i) *Balanced cold.* When the cold initial condition is distributed equally across both wells (50% left, 50% right), the crossing fraction drops to 28% (consistent with noise). The effect requires basin imbalance.

(ii) *Single-well potential.* In a harmonic potential $U = x^2 + y^2$ (no metastability), the crossing fraction is 47% (noise). The effect requires metastable basins.

(iii) *Clipping control.* Running identical simulations without gradient clipping yields 94.8% vs. 94.6% crossing, confirming the effect is not a numerical artifact.

(iv) *Seed robustness.* All 30 of 30 random seeds produce “strong” crossing ($> 70\%$ of time crossed) at the reference parameters. Mean crossing fraction: 94.7%.

The mechanism is clear: the hot distribution starts with approximately equal occupation of both wells, close to the equilibrium ratio. The cold distribution is entirely in one well and must wait for the slow barrier-crossing process to redistribute.

C. Phase diagram

Figure 2(a) shows the full phase diagram. The effect is strong ($> 90\%$ crossed) in a connected region spanning $b = 0.5\text{--}2.0$ and $T = 0.1\text{--}0.2$, with a peak of 95.4% at $b = 2.0$, $T = 0.2$. The effect vanishes in two regimes: (i) high temperature ($T > 0.5$), where thermal fluctuations wash out the basin imbalance; (ii) high barrier + low temperature ($b > 3$, $T < 0.1$), where the barrier is effectively impassable.

Twenty-five of the 72 parameter combinations show strong effects ($\geq 70\%$ of seeds with $> 70\%$ crossing).

D. Spectral prediction

Figure 2(b) shows the timescale ratio τ_1/τ_2 from the Fokker–Planck spectrum. The spectral criterion $\tau_1/\tau_2 > 10$ captures all 25 strong Mpemba points (recall = 100%). Its precision is 45%: some points with large timescale separation do not show Mpemba because the barrier is effectively impassable. This is physically correct—timescale separation is a *necessary* condition, but not sufficient.

The Kramers escape rate formula predicts λ_1 with a median accuracy ratio of 1.04, and 89% of predictions fall within a factor of 2 of the numerical value (Fig. 4). Crucially, the spectral analysis is not a post-hoc fit: it uses only the potential $U(x)$ and temperature T as inputs, with no adjustable parameters.

E. Observable-specificity explained

The eigenmodes of the Fokker–Planck operator reveal: ψ_1 (slow, inter-basin mode) is antisymmetric with one lobe in each well; ψ_2 (fast, intra-basin mode) is symmetric and localized within each well. Basin occupancy p_{left} projects strongly onto ψ_1 (it measures left-right asymmetry). Energy projects primarily onto ψ_2 and higher modes. The Mpemba crossing occurs because the hot distribution has a smaller overlap with ψ_1 than the cold distribution—but this advantage is invisible to observables that couple to faster modes.

IV. DISCUSSION

A. Relation to spectral theory and thermomajorization

Our results provide a concrete realization of the spectral Mpemba criterion of Lu and Raz [1], with the added insight of observable-specificity. Our system also provides a concrete example of the boundary of the thermomajorization framework [10]: our cold and hot initial conditions evidently do not satisfy thermomajorization ordering, since basin occupancy shows Mpemba while energy does not. In the language of Summer et al. [14], our hot initial condition has lower “athermality” in the basin-occupation sector but higher athermality in the energy sector, providing a concrete continuous-space example of the resource-theoretical hierarchy they establish for discrete systems.

B. Timescale separation as necessary but not sufficient

The spectral prediction achieves perfect recall (100%) but limited precision (45%). Large τ_1/τ_2 means the slow mode exists and is well-separated, but if τ_1 itself is astronomically large (exponentially in b/T), the redistribution simply does not occur on accessible timescales. The Mpemba effect thus requires a “Goldilocks zone” of metastability.

C. Implications for Mpemba searches

Our finding that the same system shows a strong Mpemba effect in one observable and none in another has practical implications. Experimental studies that measure only energy-like quantities may miss a Mpemba effect that is present in other observables. Conversely, claims of “no Mpemba effect” should specify which observable was measured.

D. Neural network training (negative transfer)

We also tested whether the basin-imbalance mechanism transfers to neural network training (details in the Supplemental Material). Five experimental configurations showed no crossing. The root cause is twofold: (i) the MLP weight space has no discrete metastable basins, and (ii) the “cold” initial condition was not stationary (KL drift of +7400%). This negative result is consistent with our mechanism: without metastable basins, the basin-imbalance pathway does not exist. We note that Liu and Hu [11] recently demonstrated Mpemba-like behavior in LLM training through a continuous spectral mechanism, confirming that the effect

can arise in ML without discrete basins via other pathways.

E. Limitations

(1) Our system is 2D with a specific potential; extension to higher dimensions remains open. (2) The spectral analysis uses the 1D marginal. (3) Very rare barrier crossings at high b /low T may be undersampled.

V. CONCLUSION

We have demonstrated that the Mpemba effect in a double-well Langevin system is observable-specific: it appears robustly in basin occupancy but not in energy or position variance. The mechanism is metastable basin imbalance, and the phase boundary is predicted by the Fokker–Planck spectral gap. The choice of observable determines whether the Mpemba effect is detectable.

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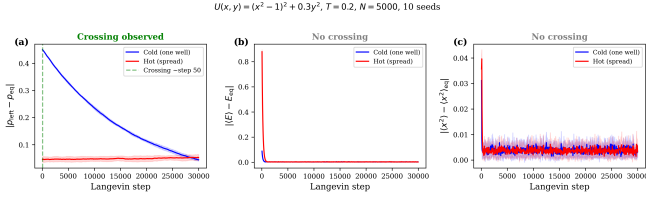


FIG. 1. Observable-specific Mpemba effect. Distance to equilibrium $|O(t) - O_{\text{eq}}|$ for cold (blue) and hot (red) initial conditions, averaged over 10 seeds. (a) Basin occupancy p_{left} shows a clear crossing. (b) Energy shows no crossing. (c) Position variance shows no crossing. Parameters: $b = 1.0$, $T = 0.2$, $N = 5000$.

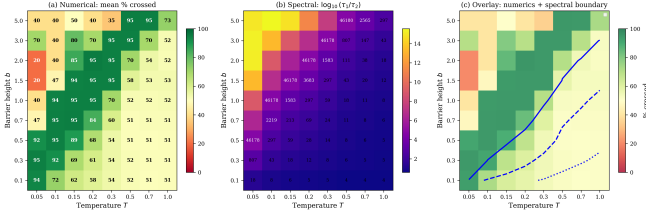


FIG. 2. Phase diagram and spectral prediction. (a) Mean % of simulation time with hot closer to equilibrium (p_{left}), across 9 barrier heights and 8 temperatures. (b) $\log_{10}(\tau_1/\tau_2)$ from Fokker-Planck spectrum. (c) Overlay with spectral contours at $\tau_1/\tau_2 = 5, 10, 50$.

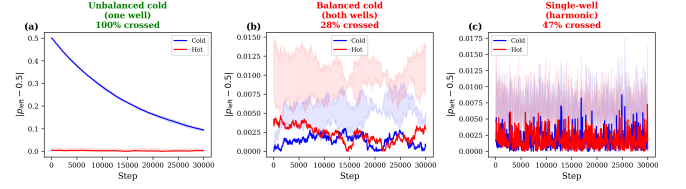


FIG. 3. Mechanism controls. (a) Unbalanced cold: strong crossing. (b) Balanced cold: effect eliminated (28%). (c) Single-well: no effect (47%).

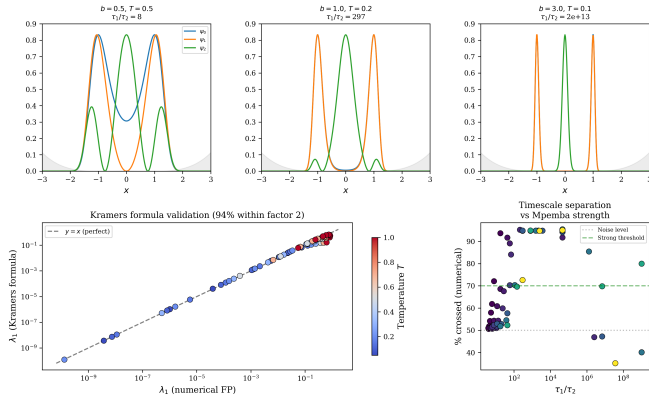


FIG. 4. Spectral analysis. Top: Fokker–Planck eigenmodes at three parameter points. Bottom left: Kramers formula vs. numerical λ_1 (94% within factor 2). Bottom right: τ_1/τ_2 vs. numerical % crossed.